

Infinitesimal Calculus 2, Moed B

2023, Reichman university.

Do not write at the back of the papers!!

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Instructions:

1. Duration: 3 hours.

Instructions:

The following exam has 5 questions. You must answer exactly 4 questions. Each question is worth 25 credits.

You must explain in full details all of your answers in all parts of the exam, and state all of the conditions for every theorem and/or claim. **Unexplained answers will not receive any credit. You do not have a choice between different items of the same question. It is forbidden to rely on any material that was not learned in the course. You must write your solutions only in the designated areas. Do not write at the back of the papers!**

1. **(25 points)** Let $x_0 \in \mathbb{R}$ and let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers.

- (a) (15 points) Let $x_1, x_2 \in \mathbb{R}$ such that $|x_2 - x_0| < |x_1 - x_0|$. Suppose that the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges for $x = x_1$. Prove that $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges absolutely for $x = x_2$.

Solution:

See lectures.

- (b) (10 points) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function that is infinitely differentiable. Let T_f be the Taylor series of f about $x_0 = 0$. Suppose that there exists an $M > 0$ such that $|f^{(n)}(x)| \leq M$ for every $n \in \mathbb{N}$ and for every $x \in \mathbb{R}$. Prove that $f(x) = T_f(x)$ for every $x \in \mathbb{R}$.

Solution: Let $x \in \mathbb{R}$. If $x = 0$, then $T_f(x) = f(0) + f'(0)x + \frac{f''(x)}{2}x^2 + \dots = f(0) + 0 + 0 + \dots = f(0)$. Suppose now that $x \neq 0$. It is sufficient to show that $\lim_{n \rightarrow \infty} R_n(x) = 0$. By the Taylor remainder formula, there exists a point c between 0 and x such that $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1}$. Thus, $|R_n(x)| = \frac{|f^{(n+1)}(c)|}{(n+1)!} |x|^{n+1} \leq M \frac{|x|^{n+1}}{(n+1)!}$. As seen in class, we have $\lim_{n \rightarrow \infty} \frac{|x|^n}{n!} = 0$. Thus by the squeeze rule, $\lim_{n \rightarrow \infty} |R_n(x)| = 0$ which implies that $\lim_{n \rightarrow \infty} R_n(x) = 0$.

2. **(25 points)** Let f be the function that is defined by $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{2n \cdot (2n+1)}$ for every $-1 \leq x \leq 1$.

- (a) (9 points) Let $x \in \mathbb{R}$. Compute the definite integral

$$\int_0^x \ln(1+t^2) dt$$

Solution: Integrating by parts we obtain

$$\begin{aligned} \int_0^x \ln(1+t^2) dt &= \left[t \ln(1+t^2) \right]_0^x - \int_0^x \frac{2t^2}{1+t^2} dt = x \ln(1+x^2) - 2 \int_0^x \frac{t^2+1-1}{1+t^2} dt \\ &= x \ln(1+x^2) - 2 \int_0^x 1 dt + 2 \int_0^x \frac{1}{1+t^2} dt = x \ln(1+x^2) - 2[t]_0^x + 2[\arctan(t)]_0^x = x \ln(1+x^2) - 2x + 2 \arctan(x) \end{aligned}$$

- (b) (6 points) Prove that f is well-defined.

Solution: We need to show that the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{2n \cdot (2n+1)}$ is convergent for every $-1 \leq x \leq 1$. Let $-1 \leq x \leq 1$. It is sufficient to prove that the series converges absolutely at x . Note that

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1} x^{2n+1}}{2n \cdot (2n+1)} \right| = \sum_{n=1}^{\infty} \frac{|x|^{2n+1}}{2n(2n+1)} \leq \sum_{n=1}^{\infty} \frac{1}{2n(2n+1)} \leq \sum_{n=1}^{\infty} \frac{1}{4n^2} = \frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

Thus, by the comparison principle, the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{2n \cdot (2n+1)}$ converges absolutely.

- (c) (10 points) Find an explicit expression for $f(x)$ for every $-1 \leq x \leq 1$.

Solution: Recall that for every $-1 < t \leq 1$ we have

$$\ln(1+t) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} t^n}{n}$$

Thus, given any $-1 \leq x \leq 1$ we get that $t = x^2$ satisfies $0 \leq t \leq 1$ and thus

$$\ln(1+x^2) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n}}{n}$$

Integrating term-by-term and dividing by 2, then for every $-1 \leq x \leq 1$ we have

$$f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{2n(2n+1)} = \frac{1}{2} \int_0^x \ln(1+t^2) dt = \frac{1}{2} x \ln(1+x^2) - x + \arctan(x)$$

3. (25 points)

- (a) (12 points) Let $a, b \in \mathbb{R}$ such that $a < b$ and let f be a function that is bounded on $[a, b]$. Let $(P_n)_{n=1}^{\infty}$ be a sequence of partitions of $[a, b]$, and let $\bar{S}(f, P_n)$ and $\underline{S}(f, P_n)$ be the upper and lower sums of f with respect to P_n for every $n \in \mathbb{N}$. Suppose that

$$\lim_{n \rightarrow \infty} [\bar{S}(f, P_n) - \underline{S}(f, P_n)] = 0$$

Prove that f is integrable on $[a, b]$.

Solution: We need to prove that $\underline{S}(f) = \bar{S}(f)$. As $\bar{S}(f) - \underline{S}(f) \geq 0$, it is sufficient to show that for every $\epsilon > 0$ we have $\bar{S}(f) - \underline{S}(f) < \epsilon$.

Let $\epsilon > 0$. As $\lim_{n \rightarrow \infty} [\bar{S}(f, P_n) - \underline{S}(f, P_n)] = 0$ then there exists an $N \in \mathbb{N}$ such that for every $n \geq N$ we have

$$|\bar{S}(f, P_n) - \underline{S}(f, P_n)| < \epsilon$$

As $\bar{S}(f) = \inf \{ \bar{S}(f, P) : P \text{ is a partition of } [a, b] \}$ then $\bar{S}(f) \leq \bar{S}(f, P_n)$ for every $n \in \mathbb{N}$. Similarly, we have $\underline{S}(f) = \sup \{ \underline{S}(f, P) : P \text{ is a partition of } [a, b] \}$, which implies that $\underline{S}(f) \geq \underline{S}(f, P_n)$ for every $n \in \mathbb{N}$. Thus,

$$\bar{S}(f) - \underline{S}(f) \leq \bar{S}(f, P_n) - \underline{S}(f, P_n) < \epsilon$$

(b) (13 points)

- i. (4 points) Compute the value of the following limit:

$$\lim_{x \rightarrow \infty} \frac{x}{\ln(1+x^2)}$$

Solution: We have, by L'hôpital's rule,

$$\lim_{x \rightarrow \infty} \frac{x}{\ln(1+x^2)} = \lim_{x \rightarrow \infty} \frac{1}{\frac{2x}{1+x^2}} = \lim_{x \rightarrow \infty} \frac{1+x^2}{2x} = \lim_{x \rightarrow \infty} \frac{2x}{2} = \lim_{x \rightarrow \infty} x = \infty$$

- ii. (9 points) Determine whether the following improper integral converges or diverges

$$\int_1^{\infty} \frac{dx}{\ln(1+x^2)}$$

Solution: As $\lim_{x \rightarrow \infty} \frac{x}{\ln(1+x^2)} = \infty$, then there exists a point $M > 0$ such that for every $x > M$ we have $\frac{x}{\ln(1+x^2)} \geq 1$, or $\frac{1}{\ln(1+x^2)} \geq \frac{1}{x}$. It follows that

$$\int_1^{\infty} \frac{dx}{\ln(1+x^2)} = \int_1^M \frac{dx}{\ln(1+x^2)} + \int_M^{\infty} \frac{dx}{\ln(1+x^2)} \geq \int_1^M \frac{dx}{\ln(1+x^2)} + \int_M^{\infty} \frac{dx}{x}$$

Note that $\int_1^M \frac{dx}{\ln(1+x^2)}$ is finite, as it is an integral of a continuous function, whereas $\int_M^{\infty} \frac{dx}{x} = [\ln(x)]_M^{\infty} = \ln(\infty) - \ln(M) = \infty$.

Thus, by the comparison principle, $\int_1^{\infty} \frac{dx}{\ln(1+x^2)} = \infty$.

4. (25 points)

- (a) (9 points) Let $0 \leq k \in \mathbb{Z}$ and let $(a_n)_{n=k}^{\infty}, (b_n)_{n=k}^{\infty}$ be two sequences of real numbers. Suppose that $a_n \geq 0$ and that $b_n > 0$ for every $n \geq k$. Suppose also that

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$$

Prove that $\sum_{n=k}^{\infty} b_n < \infty \Rightarrow \sum_{n=k}^{\infty} a_n < \infty$.

Solution: See lectures.

- (b) (16 points) Consider the sequence of real numbers $(a_n)_{n=2}^{\infty}$ that is given by $a_n = \int_n^{n+1} \frac{dx}{\ln(x)}$ for every $n \geq 2$.

- i. (7 points) Prove that $\lim_{n \rightarrow \infty} a_n = 0$.

Solution: By the Monotonicity of the definite integral, for every $n \geq 2$ we have

$$\frac{1}{\ln(n+1)} = \int_n^{n+1} \frac{dx}{\ln(n+1)} \leq \int_n^{n+1} \frac{dx}{\ln(x)} \leq \int_n^{n+1} \frac{dx}{\ln(n)} = \frac{1}{\ln(n)}$$

Thus, $\frac{1}{\ln(n+1)} \leq a_n \leq \frac{1}{\ln(n)}$ and by the squeeze rule, $\lim_{n \rightarrow \infty} a_n = 0$.

- ii. (7 points) Prove that the sequence $(a_n)_{n=2}^{\infty}$ is decreasing.

Solution: Let $n \geq 2$. We want to prove that $a_n \geq a_{n+1}$. Substituting $t = x - 1$, we obtain

$$a_{n+1} = \int_{n+1}^{n+2} \frac{dx}{\ln(x)} = \int_n^{n+1} \frac{dt}{\ln(t+1)} = \int_n^{n+1} \frac{dx}{\ln(x+1)}$$

Thus

$$a_n - a_{n+1} = \int_n^{n+1} \frac{dx}{\ln(x)} - \int_n^{n+1} \frac{dx}{\ln(x+1)} = \int_n^{n+1} \left[\frac{1}{\ln(x)} - \frac{1}{\ln(x+1)} \right] dx$$

As $\ln(x) \leq \ln(x+1)$ for every x , then $\frac{1}{\ln(x)} - \frac{1}{\ln(x+1)} \geq 0$ and thus $\int_n^{n+1} \left[\frac{1}{\ln(x)} - \frac{1}{\ln(x+1)} \right] dx \geq 0$, implying that $a_n \geq a_{n+1}$.

An alternative solution: Let $n \geq 2$. Then

$$a_n = \int_n^{n+1} \frac{dx}{\ln(x)} \geq \int_n^{n+1} \frac{dx}{\ln(n+1)} = \int_{n+1}^{n+2} \frac{dx}{\ln(n+1)} \geq \int_{n+1}^{n+2} \frac{dx}{\ln(x)} = a_{n+1}$$

- iii. (2 points) Conclude that there exists an $s \in \mathbb{R}$ such that $s = \sum_{n=2}^{\infty} (-1)^{n+1} a_n$.

Solution: As the sequence $(a_n)_{n=2}^{\infty}$ is decreasing and tends to zero, then the series $\sum_{n=2}^{\infty} (-1)^{n+1} a_n$ is a Dirichlet series and thus converges.

5. (25 points)

- (a) (2 points) Let $(x_0, y_0) \in \mathbb{R}^2$ and let F be a function that is defined at least in a neighborhood of (x_0, y_0) . Define the term: The partial derivative $F'_x(x_0, y_0)$.

Solution: The partial derivative $F'_x(x_0, y_0)$ is defined as

$$F'_x(x_0, y_0) = \lim_{x \rightarrow x_0} \frac{F(x, y_0) - F(x_0, y_0)}{x - x_0}$$

- (b) (7 points) Consider the function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ that is defined by $F(x, y) = e^x \cos(y)$ for every $x, y \in \mathbb{R}$. Compute the value of $F''_{xx}(x, y) + F''_{yy}(x, y)$ for every $x, y \in \mathbb{R}$.

Solution: We compute

$$F'_x(x, y) = e^x \cos(y), \quad F'_y(x, y) = -e^x \sin(y)$$

and

$$F'''_{xx}(x, y) = e^x \cos(y), \quad F''_{yy} = -e^x \cos(y)$$

Thus, $F''_{xx}(x, y) + F''_{yy}(x, y) = e^x \cos(y) - e^x \cos(y) = 0$.

(c) (7 points) Prove that $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{n} + \frac{1}{n+1} \right) = 1$.

Solution: Let

$$S_N = \sum_{n=1}^N (-1)^{n+1} \left(\frac{1}{n} + \frac{1}{n+1} \right), \quad \forall N \in \mathbb{N}$$

Then

$$S_{2N} = \sum_{n=1}^{2N} (-1)^{n+1} \left(\frac{1}{n} + \frac{1}{n+1} \right) = 1 + \frac{1}{2} - \frac{1}{2} - \frac{1}{3} + \frac{1}{3} + \cdots + \frac{1}{2N-1} + \frac{1}{2N} - \frac{1}{2N} - \frac{1}{2N+1} = 1 - \frac{1}{2N+1}$$

and therefore $\lim_{N \rightarrow \infty} S_{2N} = \lim_{N \rightarrow \infty} \left(1 - \frac{1}{2N+1} \right) = 1$. Note also that $S_{2N-1} - \left(\frac{1}{2N} + \frac{1}{2N+1} \right) = S_{2N}$, which implies that $S_{2N-1} = S_{2N} + \frac{1}{2N} + \frac{1}{2N+1}$. Thus $\lim_{N \rightarrow \infty} S_{2N-1} = \lim_{N \rightarrow \infty} \left(S_{2N} + \frac{1}{2N} + \frac{1}{2N+1} \right) = 1$. Thus $\lim_{N \rightarrow \infty} S_N = 1$, which implies that $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{n} + \frac{1}{n+1} \right) = 1$.

(d) (9 points) Prove that the following series is divergent

$$\sum_{n=1}^{\infty} \frac{n!3^n}{4 \cdot 7 \cdot \cdots \cdot (3n+1)}$$

Solution: We have

$$\sum_{n=1}^{\infty} \frac{n!3^n}{4 \cdot 7 \cdot \cdots \cdot (3n+1)} = \sum_{n=1}^{\infty} \frac{3 \cdot 6 \cdot \cdots \cdot 3(n-1) \cdot 3n}{4 \cdot 7 \cdot \cdots \cdot (3n+1)} \geq \sum_{n=1}^{\infty} \frac{3 \cdot 6 \cdot \cdots \cdot 3(n-1) \cdot 3n}{6 \cdot 9 \cdot \cdots \cdot (3n+3)} = \sum_{n=1}^{\infty} \frac{3}{3n+3} = \sum_{n=1}^{\infty} \frac{1}{n+1} = \sum_{n=2}^{\infty} \frac{1}{n} = \infty$$

(as any tail of the Harmonic series is divergent). Thus $\sum_{n=1}^{\infty} \frac{n!3^n}{4 \cdot 7 \cdot \cdots \cdot (3n+1)} = \infty$.